

**AMENDMENT NO. 1 JUNE 2008
TO
IS 14797 : 2000 CLASSIFICATION OF SMELTING REDUCTION
(ALTERNATE IRON MAKING) PROCESSES**

(Foreword, para 2, line 1) — Substitute '95' for '98'.

(Foreword, para 2, lines 5 and 6) — Insert 'FINEX' after 'COREX' and delete 'AISI, DIOS, AUSMELT' and insert 'PRIMUS' after 'HISMELT'.

[Page 1, clause 3, tabular matter, Sl No. (1), col 4] — Insert 'Finex' after 'AISI'.

[Page 1, clause 3, Sl No. (3), tabular matter, col 4] — Substitute 'Com' for 'com'.

[Page 2, clause 3, Sl No. (5)(b), tabular matter, col 4] — Insert 'Finex' after 'AISI'.

[Page 2, clause 3, Sl No. (6)(b), tabular matter, col 4] — Insert 'PRIMUS' after 'AISI'.

(Page 3, Annex A) — The Summary of Smelting Reduction Processes given on page 2 of this Amendment has been updated and is as follows:

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<i>Sl No.</i>	<i>Process Name / Developer(s) / Country</i>	<i>Pre-reduction Reactor (Type)</i>	<i>Smelting Reactor (Type)</i>
1)	ELRED: Stora Kopparberg AB and Asca AB, Sweden	Fluidized bed at higher pressure using smelter gas	Smelting in DC arc furnace
2)	INRED: Boliden, Sweden	Partial reduction by flash smelting	Smelting and final reduction in EAF
3)	PLASMASMELT: SKF Steel, Sweden	Pre-reduction in fluidized bed in two stages using gas from plasma furnace	Smelting in plasma furnace
4)	COREX: Korf Stahl, Germany and Voest Alpine, Austria	Pre-reduction in a shaft furnace	Melting in smelter-gasifier using lump coal and injected oxygen
5)	HISMELT: Klockner, Germany, CRA, Australia and Midrex, USA	Pre-reduction up to FeO in circulating fluid bed using smelter off gas	Smelting in smelter-gasifier
6)	COMBISMELT: Lurgi, Germany and Mannesmann Demag, Germany	Pre-reduction in rotary kiln using coal	Smelting and final reduction in EAF
7)	AISI Process: American Iron and Steel Institute and HYL, Mexico	Pre-reduction in shaft furnace using smelter off gas up to FeO stage	Molten iron bath reactor using coal and oxygen with post combustion and provision for direct steel making
8)	ROMELT (earlier FLPR— Ferrous Liquid Phase Reduction): Moscow State Institute of Steel and Alloys, Russia	Not required	Rectangular horizontal reactor with side lances and overflow type separator for slag and metal
9)	FINEX: Voest Alpine and Research Institute of Industrial Science and Technology, Korea Posco, Korea	Pre-reduction in series of fluidized bed reactors followed by compacting into briquettes; coal fines also as briquettes	Smelting in smelter-gasifier using oxygen and coal
10)	AUSMELT (AUSIRON): Ausmelt Limited, Australia	Oxygen and coal fines blown through top lance(s) into a liquid metal bath in a converter type furnace	Smelting in converter
11)	FASTMELT: Midrex, USA and Kobe Steel, Japan	Pre-reduction in Fastmet RHF	Melting in EAF
12)	ITMK3: Kobe Steel, Japan and Midrex, USA	Pre-reduction in Inmetco RHF	Melting in a SAF
13)	IDI (Redsmelt): Iron Dynamics, USA, Sumitomo, Japan and Mannesmann Demag, Germany	Pre-reduction of composite pellets of ore and coal in rotary hearth furnace	Smelting in EAF
14)	TECNORED: CAEMI, Brazil	Coal, ore, green pellets and briquettes reduced in a specially designed low shaft furnace	Melting in shaft furnace
15)	CLEANSMELT: CSM, Brazil and ILP, Italy	Reduction in Cyclone Converter Furnace	Melting in a melter-gasifier

(MTD 30)